

COM6022M - Assessment 2: Securing and Managing a Cloud-Based Network

Student ID: 250226267

June 2026

Contents

1 Task 1: Secure Cloud Network and Virtual Machines	4
1.1 Cloud-Based Network Resource Group	4
1.2 Network and Subnet Configuration.....	4
1.3 Network Security Group (NSG) Inbound Rules.....	5
1.4 The Virtual Machines	6
2 Task 2: Database Deployment and Data Security	8
2.1 Database configuration	8
2.2 Encryption at Rest	8
2.3 Encryption in Transit	9
2.4 Access Method	10
3 Task 3: Availability and Protection	10
3.1 Load Balancer	10
3.2 DDoS Protection.....	11
4 Task 4: Containerisation with Docker	12
4.0.1 Docker Installation	12
4.1 Web Application Container	13
4.2 Secure Container Communication	14
4.3 Advantages of Containerisation	14
5 Task 5: Security Validation	14
5.1 Network Security Groups (NSGs).....	14
5.2 Encryption	15
5.3 Load Balancer	15
5.4 DDoS Protection.....	15
5.5 Containers	15
6 Network Diagram and TechLinkRG Resource Visualiser	16

List of Figures

1	Cloud-Based Network Resource Group.....	4
2	TechLink-VNet overview showing address space	5
3	Application and Database subnets.....	5
4	AppNSG inbound security rules	6
5	DbNSG inbound security rules showing app-subnet as source	6
6	AppNSG subnet association.....	6
7	DbNSG subnet association.....	6
8	VM1 overview showing subnet, region, and status	7
9	VM2 overview showing subnet, region, and status	7
10	TechLinkDB overview page	8
11	TDE settings page showing status as Enabled	9
12	SQL Server connection string showing Encrypt=True parameter	9
13	TechLink-LB overview showing frontend IP and backend pool	10
14	Backend pool showing AppVM1 and AppVM2 as members	11
15	Load balancing rules showing HTTP rule and health probe.....	11
16	TechLink-VNet DDoS protection settings	12
17	Terminal on VM1 showing docker –version output	12
18	Terminal on VM2 showing docker –version output	12
19	:Docker ps output on VM1	13
20	:Docker ps output on VM2	13
21	Demonstration of techlink-container running.....	13
22	Docker network ls showing techlink-net and ping output between containers.....	14
23	Network Diagram	16
24	TechLink’s Resource Group System Visualiser	17

1 Task 1: Secure Cloud Network and Virtual Machines

1.1 Cloud-Based Network Resource Group

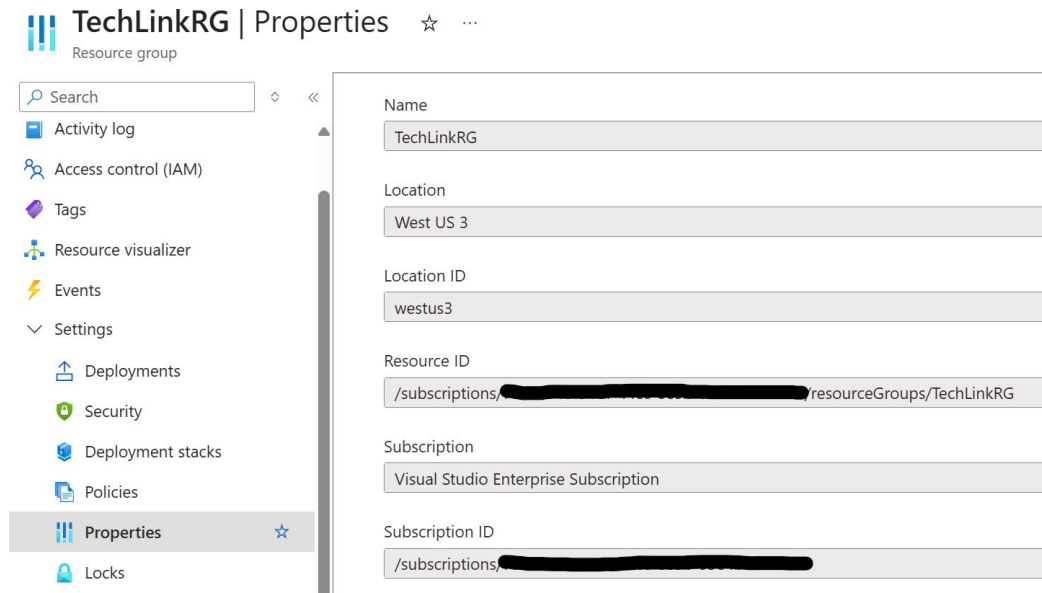


Figure 1: Cloud-Based Network Resource Group

1.2 Network and Subnet Configuration

TechLink-VNet was created in West US 3 with the address space 10.0.0.6/24. Two subnets are utilised to provide network segmentation between the application and data tiers. The application subnet address space 10.0.1.0/24 (apps-subnet) hosts the two virtual machines running the web application. The database subnet address space 10.0.2.0/24 (db-subnet) is reserved for the Azure SQL Database. The segmentation of these tiers into their separate subnets ensures the isolation of database resources from the internet, this directly addresses one of TechLink's infrastructure flaws [Microsoft(2024)].

TechLink-VNet
Virtual network

Search

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
- Monitoring
- Automation
- Help

Move Delete Refresh Give feedback

Essentials

- Resource group (move) : [TechLinkRG](#)
- Location (move) : West US 3
- Subscription (move) : [Visual Studio Enterprise Subscription](#)
- Subscription ID : [REDACTED]
- Address space : **10.0.0.0/16**
- Advertised gateway pre... : -
- Subnets : [2 subnets](#)
- DNS servers : [Azure provided DNS service](#)
- BGP community string : [Configure](#)
- Virtual network ID : 9b27d735-cef5-43ef-a133-5e215f8c18b2

Figure 2: TechLink-VNet overview showing address space

☐	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group
☐	apps-subnet	10.0.1.0/24	-	249	-	AppNSG
☐	db-subnet	10.0.2.0/24	-	251	-	DbNSG

Figure 3: Application and Database subnets

1.3 Network Security Group (NSG) Inbound Rules

Two NSG impose access control at the subnet level. AppNSG is associated with the application subnet and only permits inbound HTTP (port 80), HTTPS (port 443), and SSH (port 22) traffic. All other inbound traffic is denied by the default NSG rule. HTTP and HTTPS allow the web application to serve client requests, and SSH enables administrative access for Docker deployment. Source port ranges are left as 'any' because client source ports are temporary and unpredictable. Destination port ranges are specified.

Priority	Name	Port	Protocol	Source	Destination	Action
100	allowHTTP	80	TCP	Any	Any	Allow
110	allowHTTPS	443	TCP	Any	Any	Allow
120	allowSSH	22	TCP	Any	Any	Allow

Figure 4: AppNSG inbound security rules

DbNSG permits SQL traffic (port 1433) only from the application subnet source address range 10.0.1.0/24. All other inbound traffic is denied by a Deny-All rule with a lower priority. This configuration ensures the database is completely unreachable from the internet and can only be accessed by application-tier resources, this approach directly implements the principle of least privilege at the network layer.

Priority	Name	Port	Protocol	Source	Destination	Action
100	allow-SQL-app	1433	TCP	10.0.1.0/24	Any	Allow
200	deny-all-inbound	Any	Any	Any	Any	Deny

Figure 5: DbNSG inbound security rules showing app-subnet as source

Name	Address range	Virtual network
apps-subnet	10.0.1.0/24	TechLink-VNet

Figure 6: AppNSG subnet association

Name	Address range	Virtual network
db-subnet	10.0.2.0/24	TechLink-VNet

Figure 7: DbNSG subnet association

1.4 The Virtual Machines

VM1 and VM2 were deployed to the application subnet, running Ubuntu Server 22.04 LTS. The virtual machines are protected by AppNSG and are placed inside apps-subnet. Public IP addresses were assigned to each machine to enable SSH access for Docker installation. In a production environment, these would be replaced by Azure Bastion to eliminate direct public exposure of the management ports.

^ Essentials		JSON View
Resource group (move)	:	TechLinkRG
Status	:	Stopped (deallocated)
Location	:	West US 3 (Zone 1)
Subscription (move)	:	Visual Studio Enterprise Subscription
Subscription ID	:	[REDACTED]
Availability zone	:	1
Operating system	:	Linux
Size	:	Standard B2als v2 (2 vcpus, 4 GiB m...
Primary NIC public IP	:	20.38.38.109 1 associated public IPs
Virtual network/subnet	:	TechLink-VNet/apps-subnet
DNS name	:	Not configured

Figure 8: VM1 overview showing subnet, region, and status

^ Essentials		JSON View
Resource group (move)	:	TechLinkRG
Status	:	Stopped (deallocated)
Location	:	West US 3 (Zone 1)
Subscription (move)	:	Visual Studio Enterprise Subscription
Subscription ID	:	[REDACTED]
Availability zone	:	1
Operating system	:	Linux
Size	:	Standard B2als v2 (2 vcpus, 4 GiB m...
Primary NIC public IP	:	4.148.240.74 1 associated public IPs
Virtual network/subnet	:	TechLink-VNet/apps-subnet
DNS name	:	Not configured

Figure 9: VM2 overview showing subnet, region, and status

2 Task 2: Database Deployment and Data Security

2.1 Database configuration

TechLinkDB (Azure SQL Database) was deployed on techlink-sql-server in the West US 3 region, this is not consistent with the UK GDPR data residency requirement identified in Assessment 1 [Information Commissioner's Office(2021)]. This architectural decision was chosen according to the membership region limitation, as the membership provided does not allow for the deployment of VMs in the UK regions. Azure SQL Database was selected over Cosmos DB or PostgreSQL because TechLink's core system uses a relational database storing customer and transaction data, making SQL semantics the appropriate approach. [Information Commissioner's Office(2021)]

^ Essentials		JSON	
Resource group (...)	: TechLinkRG	Server name	: techlinksolutions-sql-server.database.windows.net
Status	: Online	Elastic pool	: No elastic pool
Location	: West US 3 	Connection strings	: Show database connection strings
Subscription (move)	: Visual Studio Enterprise Subscription	Pricing tier	: Basic

Figure 10: TechLinkDB overview page

2.2 Encryption at Rest

The Transparent Data Encryption (TDE) method is enabled by default on the systems Azure SQL Database. TDE automatically encrypts the database files, transaction logs, and backups using AES-256 without the need for any application changes [Microsoft(2023b)]. Enabling this method directly addresses TechLink's requirement to protect the customer's personal data and financial transaction records under the UK GDPR legislation and the PCI-DSS standard [Information Commissioner's Office(2021), PCI Security Standards Council(2022)].

Transparent data encryption

Always Encrypted



Transparent data encryption encrypts your databases, backups, and logs at rest without an enable encryption, go to each database. [Learn more](#)

Data encryption

ON

OFF

Encryption status

 Encrypted

Once database level customer-managed key is selected, switching to server level encryption key is or with service managed key.

Transparent data encryption ⓘ

☒ Server level encryption key
 ☐ Database level customer-managed key (CMK)

Figure 11: TDE settings page showing status as Enabled

2.3 Encryption in Transit

Azure SQL Database imposes TLS 1.2 as the minimum protocol version [Microsoft(2023b)] for all client connections. The connection string includes 'Encrypt=True' and 'TrustServerCertificate=False', these options make sure all data transmitted between the application and the database is encrypted end-to-end, preventing interception of credentials or customer data over the network.

[ADO.NET](#)
[JDBC](#)
[ODBC](#)
[PHP](#)
[Go](#)

ADO.NET (Microsoft Entra passwordless authentication)

[Microsoft.Data.SqlClient Quickstart](#)

[Entity Framework Core Quickstart](#)

```
Server=tcp:techlinksolutions-sql-server.database.windows.net,1433;Initial Catalog=TechLinkDB;Encrypt=True;TrustServerCertificate=False;Connection Timeout=30;Authentication="Active Directory Default";
```

Figure 12: SQL Server connection string showing Encrypt=True parameter

2.4 Access Method

The database was configured with the restricted access to public endpoints to Azure services only. The client IP addresses are managed via the firewall rules on the SQL Server, no public internet access beyond Azure's internal network is permitted. In a production deployment, a Private Endpoint would be configured to route all database traffic through the VNet exclusively, eliminating public exposure.

3 Task 3: Availability and Protection

3.1 Load Balancer

TechLink-LB is a Standard public load balancer which distributes inbound HTTP traffic across VM1 and VM2 using round-robin load distribution. A health probe is used to monitor port 80 on each VM at regular intervals, if a VM fails to respond, the load balancer removes automatically from the rotation and directs all traffic to the healthy instance. This approach directly addresses TechLink's documented pattern of outages caused by hardware failures, ensuring a single VM failure does not result in service unavailability. Distributing traffic across two instances provides the capacity to absorb large traffic spikes without either VM becoming a bottleneck for the system.

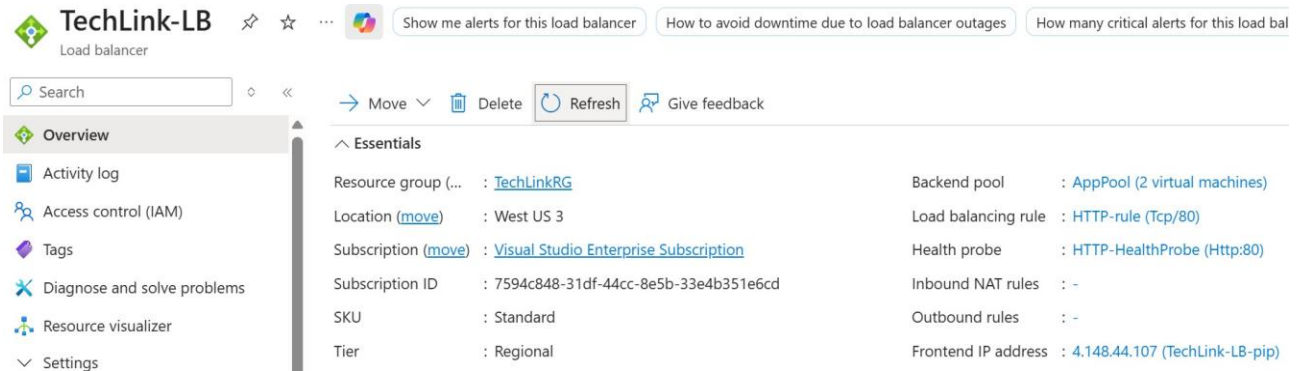


Figure 13: TechLink-LB overview showing frontend IP and backend pool

Backend pool	Resource Name	IP address	Network interf...	Availability zone	Rules count
AppPool (2)					
AppPool	VM2	10.0.1.5	vm2599_z1	1	1
AppPool	VM1	10.0.1.4	vm1419_z1	1	1

Figure 14: Backend pool showing AppVM1 and AppVM2 as members

Name	Protocol	Backend pool	Health probe
HTTP-rule	TCP/80	AppPool	HTTP-HealthProbe

Figure 15: Load balancing rules showing HTTP rule and health probe

3.2 DDoS Protection

The Azure DDoS Network Protection Standard was enabled on TechLink-VNet by means of a dedicated protection plan (TechLink-DDoS) [Microsoft(2023a)]. Unlike the default Basic tier, the DDoS Network Protection Standard delivers enhanced mitigation capabilities with adaptive tuning as protection policies are calibrated automatically against TechLink’s actual traffic baselines. Ensuring false positive cases are reduced and ensuring legitimate traffic is never dropped during an attack. This Standard protection covers all public IP resources within the virtual network, including the load balancer frontend IP and VMs, therefore defending every reachable entry point accessible externally. It provides materially stronger defence against volumetric attacks such as UDP floods, TCP SYN floods, and HTTP floods targeting the web application faced to the public.

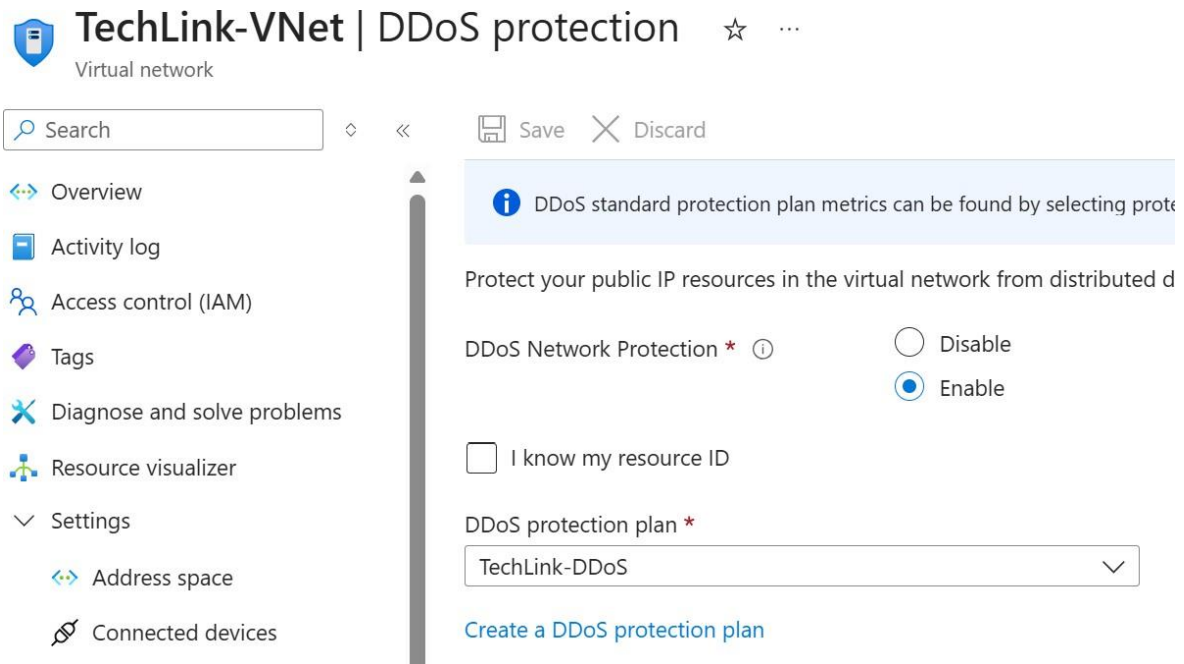


Figure 16: TechLink-VNet DDoS protection settings

4 Task 4: Containerisation with Docker

4.0.1 Docker Installation

The Docker Engine was installed on both VM1 and VM2 using SSH. The docker.io package was installed using the Ubuntu package manager (apt-get), and the Docker service was enabled to start automatically on system boot using systemctl. This was repeated identically on both virtual machines to ensure a consistent deployment environment across the application tier.

```
azureuser@VM1:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu3~24.04.2
```

Figure 17: Terminal on VM1 showing docker --version output

```
azureuser@VM2:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu3~24.04.2
```

Figure 18: Terminal on VM2 showing docker --version output

4.1 Web Application Container

A custom Docker image (techlink-webapp) was built from an nginx:alpine base image, creating a HTML page representing the TechLink web application. The container was launched with port 80 mapped to the host, making the application accessible by the VM's public IP address and through the load balancer.

```
azureuser@VM1:~/techlink-app$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
6ce606e48854   techlink-webapp "/docker-entrypoint..." 7 seconds ago  Up 7 seconds  0.0.0.0:80->80/tcp, [::]:80->80/tcp techlink-container
```

Figure 19: :Docker ps output on VM1

```
azureuser@VM2:~/techlink-app$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
a105be1e1719   techlink-webapp "/docker-entrypoint..." 6 seconds ago  Up 5 seconds  0.0.0.0:80->80/tcp, [::]:80->80/tcp techlink-container2
```

Figure 20: :Docker ps output on VM2

```
azureuser@VM1:~$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:0e760fd9bc48ba8041e7c6db999bb40bfca508b4be580ac75d32c4e29d202ce1
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

Figure 21: Demonstration of techlink-container running

4.2 Secure Container Communication

A custom Docker bridge network (techlink-net) was created on each VM. Containers linked to the same bridge network can communicate with each other by container names, this facilitates internal DNS resolution within the network. Containers on different networks remain isolated from each other, implementing network segmentation at the container level. The communication between the containerised application and the Azure SQL Database is secured with the use of TLS, enforced by the connection string configured with 'Encrypt=True'.

```
^Cazureuser@VM1:~$ docker run --rm --network techlink-net alpine ping -c 4 techlink-container2
PING techlink-container2 (172.18.0.2): 56 data bytes
64 bytes from 172.18.0.2: seq=0 ttl=64 time=0.079 ms
64 bytes from 172.18.0.2: seq=1 ttl=64 time=0.052 ms
64 bytes from 172.18.0.2: seq=2 ttl=64 time=0.050 ms
64 bytes from 172.18.0.2: seq=3 ttl=64 time=0.072 ms
```

Figure 22: Docker network is showing techlink-net and ping output between containers

4.3 Advantages of Containerisation

The primary advantage is scalability. Containers are lightweight and start in seconds, allowing a swift horizontal scaling when traffic increases; new container instances can be launched on additional VMs without the overhead of provisioning a full VM. This directly addresses one of TechLink's documented vulnerability regarding unpredictable traffic spikes.

The second advantage, is consistency and portability: the Docker image packages the application and all its dependencies together, thereby guaranteeing that the application behaves identically across VM1 and VM2 regardless of any differences in the underlying operating system configuration. This eliminates 'it works on my machine' failures and simplifies deployment during the migration phases defined in the Assessment 1 roadmap.

5 Task 5: Security Validation

The following section summarises how each deployed security control protects TechLink's cloud environment.

5.1 Network Security Groups (NSGs)

The NSGs act as stateful virtual firewalls applied at the subnet level, filtering all inbound and outbound traffic before it reaches any resource. AppNSG restricts the application subnet to HTTP, HTTPS, and SSH only, reducing the attacking opportunities by blocking all other entry points. DbNSG ensures the

database subnet is completely isolated and detached from the internet, accepting SQL connections only from the known application subnet address range. They enforce the VNet segmentation designed in Assessment 1 and prevent lateral movement between network zones.

5.2 Encryption

TDE ensures customer and transaction data stored in Azure SQL Database is protected against unauthorised physical access or backup theft by means of encrypting data files at rest using AES-256 [Microsoft(2023b)]. TLS 1.2 protects data in transit between the application containers and the database, thereby preventing interception or tampering over the network. The controls satisfy UK GDPR Article 32 and PCI-DSS Requirement 3 and 4 obligations to protect personal data and cardholder data both at rest and in transit [Information Commissioner's Office(2021), PCI Security Standards Council(2022)]..

5.3 Load Balancer

The Standard load balancer enhances availability by the distribution of client requests across both application VMs. Health probes detect VM or application failures within seconds and remove unhealthy instances from rotation automatically, maintaining service continuity without manual intervention. This helps mitigate TechLink's historical single-point-of-failure risk, where a single hardware failure caused a full outage.

5.4 DDoS Protection

Azure DDoS Network Protection Standard, deployed using the TechLink-DDoS protection plan. This plan facilitates adaptive mitigation of volumetric attacks across all public IPs within TechLink-VNet (load balancer frontend and both virtual machines). Protection policies are automatically tuned to TechLink's traffic baselines, which reduces false positives. This Standard tier delivers significantly stronger defence than the default Basic tier against volumetric threats such as UDP floods and TCP SYN floods [Microsoft(2023a)].

5.5 Containers

The Docker containers supply process-level isolation between the application and the host operating system, this has a direct impact in limiting the blast radius of any application level vulnerability. The custom bridge network (techlink-net) restricts container-to-container communication to connected containers explicitly, thereby preventing arbitrary inter-container access. Container images cannot be altered and are version-controlled, enabling consistent, auditable deployments and fast rollback if a vulnerability is discovered in a deployed image.

6 Network Diagram and TechLinkRG Resource Visualiser

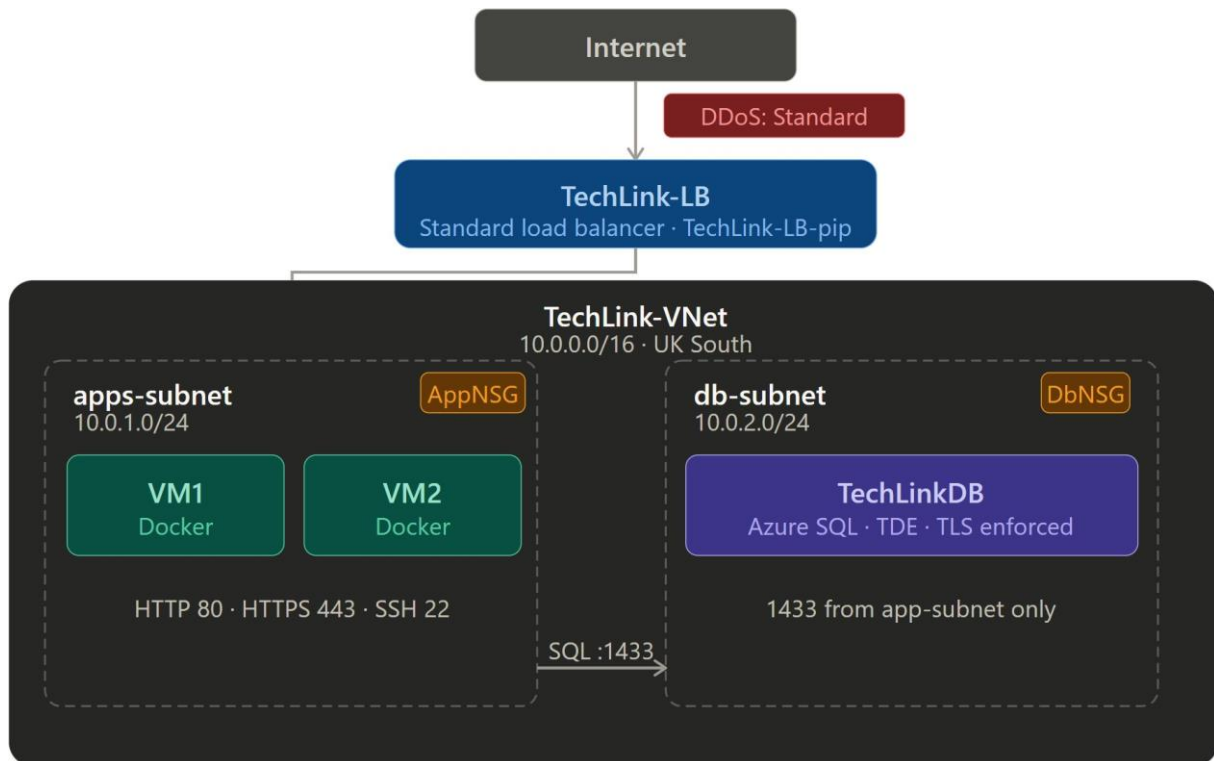


Figure 23: Network Diagram

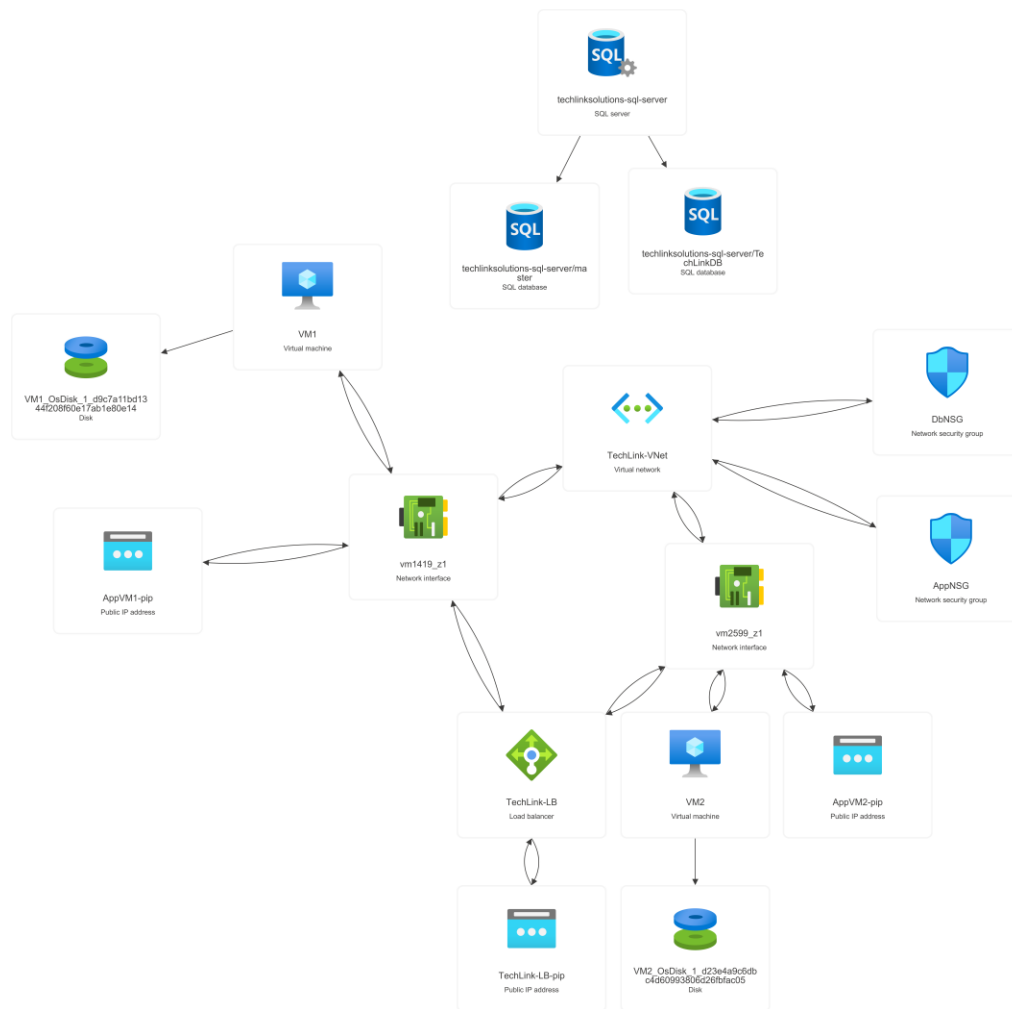


Figure 24: TechLink's Resource Group System Visualiser

References

- [Information Commissioner's Office(2021)] Information Commissioner's Office. 2021. Guide to the UK GDPR. <https://ico.org.uk/for-organisations/guide-to-data-protection/guide-to-the-general-data-protection-regulation-gdpr/>.
- [Microsoft(2023a)] Microsoft. 2023a. Azure DDoS Protection Overview. <https://learn.microsoft.com/en-us/azure/ddos-protection/ddos-protection-overview>.
- [Microsoft(2023b)] Microsoft. 2023b. Transparent Data Encryption for SQL Database. <https://learn.microsoft.com/en-us/azure/azure-sql/database/transparent-data-encryption-tde-overview>.
- [Microsoft(2024)] Microsoft. 2024. Microsoft Cloud Adoption Framework for Azure. <https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/>.
- [PCI Security Standards Council(2022)] PCI Security Standards Council. 2022. PCI DSS v4.0. https://www.pcisecuritystandards.org/document_library/.